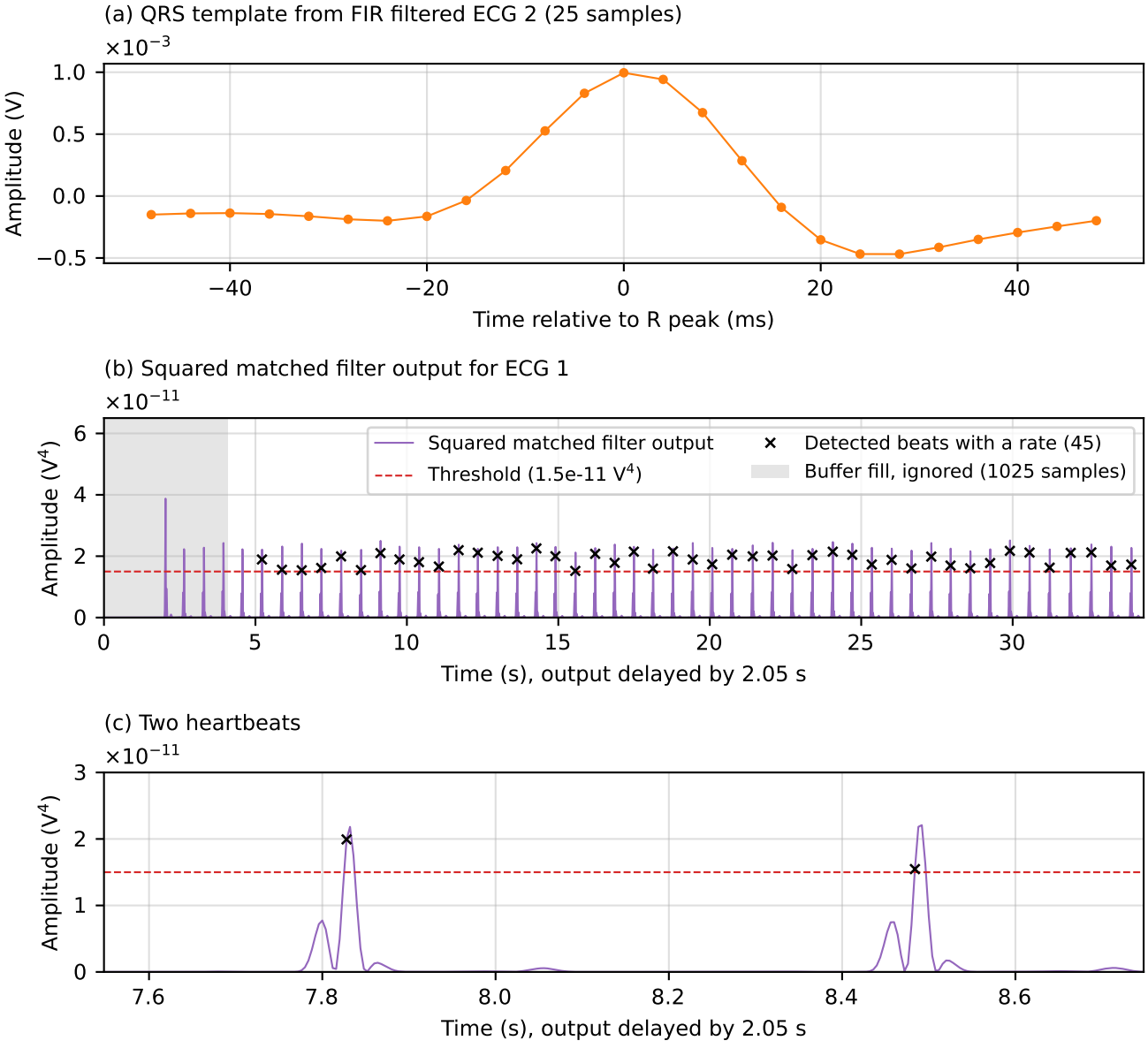
(a) QRS template from FIR filtered ECG 2 (25 samples)
×10⁻³
Amplitude (V)
1.0
0.5
0.0
−0.5
−40
−20
0
20
40
Time relative to R peak (ms)
(b) Squared matched filter output for ECG 1
×10⁻¹¹
Amplitude (V⁴)
Squared matched filter output
Threshold (1.5e-11 V⁴)
Detected beats with a rate (45)
Buffer fill, ignored (1025 samples)
6
4
2
0
0
5
10
15
20
25
30
Time (s), output delayed by 2.05 s
(c) Two heartbeats
×10⁻¹¹
Amplitude (V⁴)
3
2
1
0
7.6
7.8
8.0
8.2
8.4
8.6
Time (s), output delayed by 2.05 s